

Photon: Efficient Prefix-Conditioned Image Captioning with Lightweight Transformer Decoding

Lakshmi Ganapathi Kodi^{1*}, Dharmendra Chauhan²,
Dhanush Reddy Gangireddy¹, Sonam Kumari Chaudhary¹,
Kalidas Yeturu^{2*}, Nikhil Srivastava¹

¹National Institute of Technology, Mizoram, Aizawl, India.

²Indian Institute of Technology Tirupati, Tirumala – Tirupati, India.

*Corresponding author(s). E-mail(s): bt23ma005@nitmz.ac.in;
ykalidas@iittp.ac.in;

Contributing authors: cs24m115@iittp.ac.in; bt23ma003@nitmz.ac.in;
bt23ma009@nitmz.ac.in; nikhil.math@nitmz.ac.in;

Abstract

Image captioning systems based on large vision-language models often rely on computationally intensive cross-attention architectures, limiting their suitability for real-time and resource-constrained deployment. In this work, we propose *Photon*, a lightweight prefix-conditioned multimodal captioning framework that combines a frozen MobileCLIP vision encoder with a compact decoder-only Transformer incorporating Rotary Positional Embeddings, RMSNorm, and SwiGLU activations. Visual information is injected through a small set of learned prefix tokens, enabling efficient multimodal conditioning without region-based processing or heavy cross-modal attention. On the MS-COCO Karpathy split, Photon achieves a CIDEr score of 108.59 while requiring only 12.41M trainable parameters and 3.72 GFLOPs for image to caption generation. The model demonstrates competitive semantic alignment performance and improves inference efficiency, achieving $2.41\times$ GPU and $8.28\times$ CPU speed-ups compared to larger pretrained baselines. Zero-shot evaluation on Flickr30K, NoCaps and TextCaps further indicates consistent cross-dataset generalization across lexical, consensus-based, and embedding-based metrics. Batch-scaling analysis reveals near-linear throughput growth up to batch size 512, highlighting effective parallelization of the decoder. These results suggest that prefix-based multimodal conditioning with a modern lightweight decoder provides a favorable balance

between caption quality and computational efficiency, making it suitable for practical deployment scenarios.

Keywords: Image Captioning, Multimodal Learning, Prefix Conditioning, Vision–Language Models, Transformer Decoding

1 Introduction

Image captioning aims to generate natural language descriptions of visual scenes and has become a fundamental task in multimodal learning. Recent advances in vision–language pretraining have substantially improved caption quality by leveraging large-scale datasets and powerful cross-modal attention architectures. However, many high-performing systems rely on computationally intensive region-based processing or deep cross-attention modules, resulting in substantial parameter counts, high FLOPs, and limited suitability for real-time or resource-constrained deployment. Many real-world applications such as mobile accessibility systems, robotic perception, and edge-AI assistants require caption generation under strict computational budgets.

This creates a practical gap between captioning quality and deployment efficiency. While large vision–language models achieve strong performance, their computational requirements hinder applications in edge devices, embedded systems, and latency-sensitive environments. An important research question therefore arises: *can high-quality caption generation be achieved with substantially reduced architectural complexity?*

Recent prefix-based conditioning strategies suggest that visual information can be injected into language models without explicit cross-attention layers. Instead of fusing modalities through heavy interaction blocks, visual embeddings can be projected into a small set of learnable prefix tokens that condition autoregressive decoding. This paradigm offers a promising direction for reducing computational overhead while preserving multimodal grounding.

Unlike prior prefix-based methods such as ClipCap [1] that rely on large pre-trained decoders, Photon introduces a lightweight decoder-only architecture that removes cross-attention entirely and replaces fine-grained multimodal fusion with efficient global prefix conditioning. This represents a shift from token-level alignment to compact semantic control for caption generation.

Motivated by this observation, we introduce *Photon*, a lightweight prefix-conditioned multimodal captioning framework. Photon combines a frozen MobileCLIP visual encoder with a compact decoder-only Transformer incorporating Rotary Positional Embeddings [2], RMSNorm, and SwiGLU [3] activations. Visual embeddings are projected into a fixed number of prefix tokens that condition the decoder without region-level processing or additional cross-attention layers.

Extensive experiments demonstrate that Photon achieves competitive caption quality on the MS-COCO Karpathy split while maintaining low computational complexity. The model uses 12.41M trainable parameters and requires 3.72 GFLOPs, while also

providing substantial GPU and CPU inference speed-ups compared to larger pre-trained baselines. Additional analyses on cross-dataset generalization and batch-wise scaling further highlight the deployment suitability of the proposed design.

Design Principle.

Our approach is motivated by the observation that effective visual–language alignment for caption generation does not necessarily require heavy cross-attention or region-level processing. Instead, a compact language model can be conditioned through a small number of visual prefix tokens that encode global scene semantics. This design enables the decoder to reuse its language modeling capacity while receiving sufficient visual context for caption generation. The resulting architecture separates visual feature extraction from language generation while maintaining efficient information transfer between modalities. This principle allows the model to achieve strong caption quality while significantly reducing parameter count, computational cost, and inference latency.

The main contributions of this work are summarized as follows:

- We propose a lightweight prefix-conditioned captioning architecture that avoids heavy cross-modal attention while preserving strong visual grounding.
- We demonstrate that a compact decoder with modern architectural components can achieve competitive performance with significantly reduced computational cost.
- We provide a comprehensive quality–efficiency analysis, including parameter/FLOP comparisons, inference speed evaluation, cross-dataset generalization, and batch-scaling behavior.

2 Related Work

Image captioning has evolved from region-based attention models to large-scale vision–language pretraining frameworks. Early influential approaches such as the Bottom-Up and Top-Down attention model (Up-Down) [4] leveraged region-level features extracted using object detectors (e.g., Faster R-CNN) and applied attention-based decoding over detected object proposals. This paradigm explicitly models object-centric representations but introduces additional computational overhead due to region extraction and cross-modal attention.

More recent approaches build upon pretrained vision–language encoders such as CLIP. ClipCap [1] projects a global CLIP image embedding into a sequence of prefix tokens that condition a pretrained GPT-2 language model. This design eliminates explicit cross-attention but relies on a large autoregressive backbone for generation. CapDec [5] further explores decoding from CLIP embeddings by training a language model to reconstruct text from image representations, often incorporating text-only objectives. While effective, these approaches primarily depend on large pretrained decoders. ViT-GPT2 [6] follows a similar paradigm by combining a Vision Transformer (ViT) encoder with a GPT-2 decoder, where visual features are projected into the language embedding space for autoregressive caption generation without explicit cross-attention modules. While effective, these approaches primarily depend on large pretrained decoders.

Retrieval-augmented captioning methods such as SmallCap [7] incorporate externally retrieved captions and employ cross-attention between CLIP features and a GPT-based decoder. Although retrieval improves caption diversity and grounding, it increases inference complexity and memory requirements.

Large-scale vision-language pretraining frameworks, such as BLIP [8], combine multimodal pretraining objectives with cross-attention architectures to achieve strong captioning performance. Similarly, region-based vision-language pretraining models such as VinVL [9] and OSCAR [10] employ object-centric representations fused through transformer-based multimodal pretraining. These models achieve competitive results but involve substantial computational cost and large parameter counts.

In contrast to region-based or cross-attention-heavy architectures, the proposed approach adopts a prefix-conditioned design that injects global visual embeddings directly into a compact decoder-only Transformer. By avoiding region detection and additional cross-modal attention layers, the architecture emphasizes computational efficiency while maintaining competitive caption quality. These observations motivate the exploration of compact prefix-conditioned architectures that balance caption quality and computational efficiency.

Recent advances in multimodal learning have increasingly emphasized data-driven modeling, uncertainty-aware learning, and distributed training paradigms. Uncertainty-aware approaches aim to quantify prediction confidence and improve robustness, particularly in complex multimodal settings, while distributed and federated learning frameworks enable scalable training across decentralized data sources. Representative works in this direction include recent data-driven approaches and federated Bayesian learning methods for uncertainty quantification and scalable multimodal systems [11–14]. In contrast to these trends, the proposed Photon framework focuses on architectural efficiency and compact design, demonstrating that competitive captioning performance can be achieved without relying on large-scale or computationally intensive training strategies. This highlights a complementary perspective, where lightweight prefix-conditioned models can serve as efficient alternatives within the broader landscape of robust and scalable multimodal systems.

3 Methodology

This section describes the overall architecture (Figure 1), the end-to-end training pipeline (Figure 2), the mathematical formulation of prefix conditioning and the training objective (Section 3.3), and the algorithmic procedures for training and inference (Algorithms 1A and 1B).

Overview. Photon is a lightweight prefix-conditioned image captioning model consisting of three components: (1) a frozen MobileCLIP encoder, (2) a prefix projection module generating $K = 8$ visual tokens, and (3) a compact decoder-only Transformer. Unlike encoder-decoder architectures, Photon performs multimodal fusion using unified self-attention over concatenated visual and textual tokens.

3.1 Model Architecture

Figure 1 presents the overall architecture of the proposed Photon model. The framework consists of three main components: a frozen MobileCLIP-S1 [15] visual encoder, which employs a FastViT [16] backbone, a prefix projection module, and a compact decoder-only Transformer.

The MobileCLIP-S1 encoder extracts a 512-dimensional global visual embedding from the input image. This embedding is L2-normalized and passed to a two-layer MLP projection module, which expands the visual representation into $K = 8$ prefix tokens. These prefix tokens are concatenated with text token embeddings and processed by a 6-layer decoder-only Transformer [17] based on a Nano-LLaMA style design.

Unlike conventional encoder-decoder architectures that rely on explicit cross-attention layers, our approach performs multimodal interaction through a unified self-attention mechanism applied to the concatenated sequence of visual prefix tokens and textual tokens. This formulation allows textual tokens to directly attend to visual context within the same attention operation, enabling effective cross-modal information flow without introducing additional cross-attention modules. As a result, the model reduces architectural complexity and computational overhead while maintaining strong visual grounding. We note that while cross-modal interactions are still enabled through shared self-attention over the combined sequence, our approach eliminates the need for dedicated cross-attention layers used in conventional encoder-decoder architectures.

From a theoretical perspective, prefix tokens act as a global conditioning signal in autoregressive Transformers, where early tokens influence all subsequent predictions through self-attention. This enables prefix conditioning to provide a compact semantic representation of the visual input, guiding generation without requiring explicit token-level alignment. For captioning tasks that primarily depend on global scene understanding, such conditioning is sufficient while being computationally more efficient.

Each decoder layer consists of RMSNorm [18], RoPE-based causal self-attention, and a SwiGLU [3] feed-forward network. Self-attention is implemented using PyTorch’s scaled dot-product attention, which utilizes FlashAttention-style memory-efficient kernels when available [19]. The language modeling head is weight-tied with the input embedding layer.

3.2 Training Pipeline

Figure 2 illustrates the end-to-end training pipeline. Given an input image and its ground-truth caption, the image is first encoded into a normalized embedding. The embedding is projected into prefix tokens and concatenated with caption token embeddings, and the combined sequence is processed by the Transformer decoder.

During training, teacher forcing [20] is applied. The decoder predicts the next token at each step, and the predicted logits are shifted and compared against the shifted ground-truth tokens using cross-entropy loss.

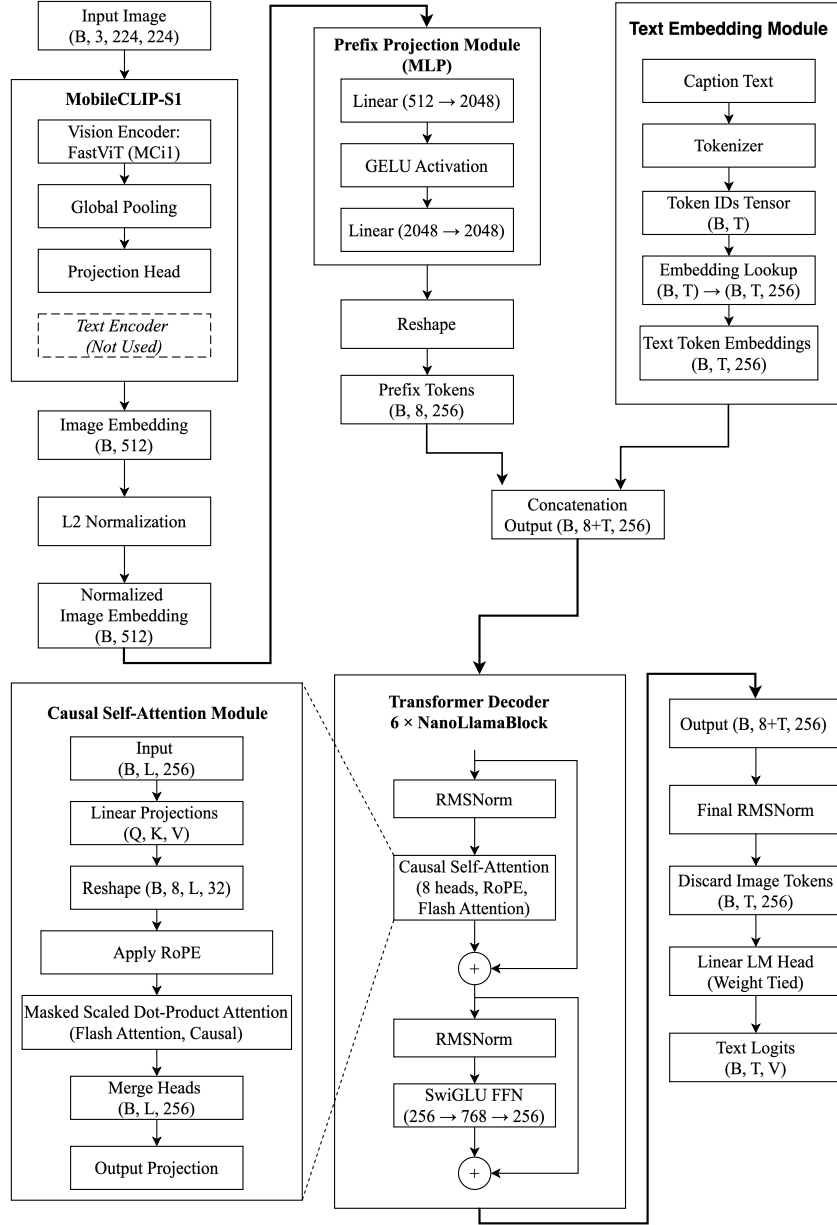


Fig. 1: Overall architecture of the proposed Photon model. The frozen MobileCLIP encoder extracts a global visual embedding, which is projected into prefix tokens and concatenated with text embeddings before being processed by a 6-layer Transformer decoder.

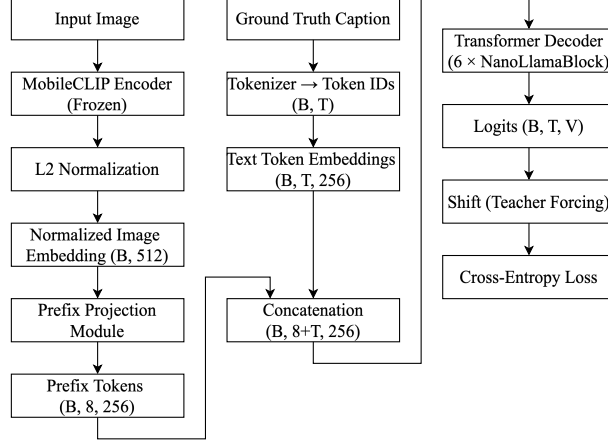


Fig. 2: End-to-end training pipeline. The image embedding is projected into prefix tokens, concatenated with token embeddings, and processed by the Transformer decoder. Training is performed using teacher-forced next-token prediction with cross-entropy loss.

3.3 Prefix Conditioning and Training Objective

Given an input image I , a frozen MobileCLIP encoder extracts a global visual embedding:

$$\mathbf{v} = f_{\text{CLIP}}(I), \quad \mathbf{v} \in \mathbb{R}^{512}. \quad (1)$$

The embedding is L_2 normalized:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}. \quad (2)$$

A two-layer projection network transforms the normalized embedding into K visual prefix tokens. The projection is implemented as a multilayer perceptron:

$$\mathbf{z} = W_2 \sigma(W_1 \hat{\mathbf{v}}), \quad (3)$$

where $\sigma(\cdot)$ denotes the GELU activation function,

$$W_1 \in \mathbb{R}^{(Kd) \times 512}, \quad W_2 \in \mathbb{R}^{(Kd) \times (Kd)},$$

$d = 256$ is the decoder hidden dimension and $K = 8$ is the number of visual tokens. The projected vector is reshaped to form prefix tokens:

$$\mathbf{P} = \text{reshape}(\mathbf{z}) \in \mathbb{R}^{K \times d}. \quad (4)$$

Let a caption be represented as a token sequence

$$\mathbf{y} = (y_1, y_2, \dots, y_T). \quad (5)$$

Let $W_{\text{emb}} \in \mathbb{R}^{V \times d}$ denote the learnable token embedding matrix.

Each token is mapped to an embedding using a learnable embedding matrix:

$$E = \text{Embed}(y; W_{\text{emb}}) \in \mathbb{R}^{T \times d}, \quad (6)$$

where $W_{\text{emb}} \in \mathbb{R}^{V \times d}$ denotes the learnable token embedding matrix.

The visual prefix tokens and textual embeddings are concatenated to form the decoder input:

$$\mathbf{X}_0 = [\mathbf{P}; \mathbf{E}] \in \mathbb{R}^{(K+T) \times d}. \quad (7)$$

The concatenated sequence is processed by a Transformer decoder composed of L layers using RMSNorm, multi-head self-attention with rotary positional embeddings (RoPE), and SwiGLU feed-forward networks:

$$\mathbf{O} = D_\theta(\mathbf{X}_0), \quad \mathbf{O} \in \mathbb{R}^{(K+T) \times d}. \quad (8)$$

The first K outputs correspond to visual prefix tokens and are discarded. The remaining outputs correspond to textual tokens:

$$\mathbf{O}^{\text{text}} = (o_1, \dots, o_T) \in \mathbb{R}^{T \times d}. \quad (9)$$

Each decoder output is mapped to vocabulary logits using the language modeling head:

$$l_t = W_{\text{emb}} o_t, \quad (10)$$

where $o_t \in \mathbb{R}^d$ and $l_t \in \mathbb{R}^V$. Thus, the model uses weight tying between the input embedding matrix and the output projection matrix.

The probability distribution over the next token is

$$q_t = \text{Softmax}(l_t) \quad (11)$$

$$q_t(v) = P(v|y_{\leq t}, I)$$

The cross entropy equation becomes,

$$\mathcal{L}_{CE} = - \sum_{t=1}^{T-1} \log(q_t(y_{t+1})) \quad (12)$$

In practice, this objective is implemented by shifting the predicted logits and ground-truth tokens by one position and computing cross-entropy between the predicted distribution and the ground-truth next token.

3.4 Training Procedure

Algorithm 1A summarizes the training process. The image embedding is extracted and normalized, projected into prefix tokens, concatenated with caption embeddings, and processed by the Transformer decoder. The first K outputs are discarded, and the remaining outputs are used to compute the next-token prediction loss.

Algorithm 1A Training Procedure

Require: Image I , ground-truth caption $\mathbf{y}$ **Require:** Frozen encoder f_{CLIP} , projection network g_θ , decoder D_θ **Ensure:** Updated trainable parameters θ

- 1: $\mathbf{v} \leftarrow f_{\text{CLIP}}(I)$
 - 2: $\hat{\mathbf{v}} \leftarrow \mathbf{v} / \|\mathbf{v}\|_2$
 - 3: $\mathbf{z} \leftarrow g_\theta(\hat{\mathbf{v}})$
 - 4: $\mathbf{P} \leftarrow \text{reshape}(\mathbf{z})$
 - 5: $\mathbf{E} \leftarrow \text{Embed}(\mathbf{y})$
 - 6: $\mathbf{X}_0 \leftarrow [\mathbf{P}; \mathbf{E}]$
 - 7: $\mathbf{O} \leftarrow D_\theta(\mathbf{X}_0)$
 - 8: Discard first K prefix outputs
 - 9: $\mathbf{O}^{\text{text}} \leftarrow (o_1, \dots, o_T)$
 - 10: Compute shifted logits $\mathbf{O}_{1:T-1}^{\text{text}}$ and labels $\mathbf{y}_{2:T}$
 - 11: $\mathcal{L} \leftarrow \text{CrossEntropy}(\mathbf{O}_{1:T-1}^{\text{text}}, \mathbf{y}_{2:T})$
 - 12: Update θ using AdamW
-

Algorithm 1B Inference Procedure (Greedy Decoding)

Require: Image I , maximum length T_{max} **Ensure:** Generated caption $\hat{\mathbf{y}}$

- 1: $\mathbf{v} \leftarrow f_{\text{CLIP}}(I)$
 - 2: $\hat{\mathbf{v}} \leftarrow \mathbf{v} / \|\mathbf{v}\|_2$
 - 3: $\mathbf{z} \leftarrow g_\theta(\hat{\mathbf{v}})$
 - 4: $\mathbf{P} \leftarrow \text{reshape}(\mathbf{z})$
 - 5: Initialize $\hat{\mathbf{y}} \leftarrow [\text{CLS}]$
 - 6: **for** $t = 1$ to T_{max} **do**
 - 7: $\mathbf{E} \leftarrow \text{Embed}(\hat{\mathbf{y}})$
 - 8: $\mathbf{X}_0 \leftarrow [\mathbf{P}; \mathbf{E}]$
 - 9: $\mathbf{O} \leftarrow D_\theta(\mathbf{X}_0)$
 - 10: Discard first K prefix outputs
 - 11: $y_{t+1} \leftarrow \arg \max_y P(y \mid \hat{\mathbf{y}}_{\leq t}, I)$
 - 12: Append y_{t+1} to $\hat{\mathbf{y}}$
 - 13: **if** y_{t+1} is [SEP] **then**
 - 14: **break**
 - 15: **end if**
 - 16: **end for**
 - 17: **return** $\hat{\mathbf{y}}$
-

3.5 Inference Procedure

Algorithm 1B describes the greedy decoding process used during inference. Caption generation begins with the [CLS] token. At each step, the decoder predicts the probability distribution over the vocabulary conditioned on the previously generated tokens and the image prefix. The token with the highest probability is selected using the argmax operation and appended to the generated sequence. Decoding stops when the [SEP] token is generated or when the maximum length T_{max} is reached.

4 Experimental Setup

This section describes the datasets, implementation details, evaluation metrics and baseline methods used to assess the proposed Photon framework.

4.1 Dataset

All experiments are conducted on the MS-COCO [21] 2014 dataset using the Karpathy split [22]. The dataset contains 113,287 training images, 5,000 validation images, and 5,000 test images. Each image is annotated with five human-written captions.

MS-COCO images vary in resolution and aspect ratio, typically ranging from approximately 200×200 to over 600×600 pixels. Each image is resized so that the

shorter side equals 224 pixels while preserving aspect ratio, followed by a center crop of size 224×224 . The cropped image is converted to RGB and normalized. The resulting tensor of shape $3 \times 224 \times 224$ is fed to the frozen MobileCLIP-S1 encoder to obtain a 512-dimensional embedding, which is L2-normalized before prefix projection.

Captions are tokenized using a Byte-Pair Encoding (BPE) [23] tokenizer trained exclusively on the MS-COCO Karpathy training split to avoid linguistic leakage. The tokenizer uses a vocabulary size of 8,000, minimum frequency of 2, and byte-level pre-tokenization. Special tokens include [UNK], [CLS], [SEP], [PAD], and [MASK]. Captions are formatted as [CLS] caption [SEP], truncated to 32 tokens, and padded when necessary.

For cross-dataset generalization analysis, we additionally evaluate the trained model on Flickr30K [24], NoCaps [25], and TextCaps [26] without further fine-tuning. Detailed results are presented in Section 5.4.

4.2 Implementation Details

The proposed architecture consists of a 6-layer decoder-only Transformer with a hidden dimension of 256 and 8 self-attention heads, corresponding to 32 dimensions per head. The feed-forward network employs SwiGLU activation with an intermediate dimension of 768. A total of 8 visual prefix tokens are generated from the 512-dimensional MobileCLIP-S1 [15] embedding using a two-layer MLP projector. The language modeling head is weight-tied with the token embedding layer. Captions are tokenized using a Byte-Pair Encoding (BPE) [23] tokenizer with a vocabulary size of 8,000 and a minimum token frequency of 2. The maximum caption length is fixed to 32 tokens. The MobileCLIP-S1 backbone produces a 512-dimensional image embedding that is L2-normalized prior to prefix projection.

The model is optimized using AdamW [27] with a learning rate of 3×10^{-4} and weight decay of 1×10^{-2} . Training is conducted for 10 epochs with a batch size of 512. No learning rate scheduler or gradient clipping is applied. The objective function is next-token cross-entropy loss under teacher forcing. The checkpoint with the lowest validation loss is selected for evaluation. Mixed-precision training is employed using automatic mixed precision, where computations are performed in FP16 while master weights are maintained in FP32.

All experiments were conducted on a Linux-based system (Kernel 5.15.0-157-generic) equipped with an NVIDIA GeForce RTX 2080 Ti GPU with 11 GB VRAM. The system includes an Intel Xeon E5-2620 v4 CPU @ 2.10 GHz (32 cores) and 125.76 GB of system RAM. The deep learning framework used was PyTorch 2.10.0 with CUDA 12.8 support, and GPU acceleration was enabled for all training and inference tasks. Precomputed image embeddings are used to reduce computational overhead during training. A complete 10-epoch training run requires approximately 16 minutes.

At inference time, greedy decoding is used. Generation begins with the [CLS] token and proceeds autoregressively. Decoding stops when the [SEP] token is generated or when the maximum length of 32 tokens is reached.

4.3 Evaluation Metrics

We report standard captioning metrics described in Table 1 including: (a) BLEU-1 to BLEU-4 (B@1–B@4); (b) METEOR; (c) ROUGE-L; (d) CIDEr; (e) SPICE.

To assess semantic alignment, we additionally report: (a) BERTScore; (b) SBERT similarity; (c) CLIPScore.

All metrics are computed using the official MS-COCO evaluation toolkit where applicable.

Table 1: Comprehensive Summary of Evaluation Metrics

Metric	Category	Description
BLEU-1 to 4 [28]	N-gram Precision	Evaluates n-gram precision between generated and multiple reference captions.
CIDEr [29]	Consensus-based	Computes TF-IDF weighted n-gram similarity across reference captions.
METEOR [30]	Alignment-based	Measures unigram alignment using synonym and stemming matching.
ROUGE-L [31]	Sequence-based	Computes longest common subsequence to assess structural similarity.
SPICE [32]	Semantic-based	Compares scene-graph tuples to evaluate semantic content agreement.
BERTScore [33]	Contextual Sim.	Computes token-level similarity using contextual BERT embeddings.
Rescaled BS [33]	Normalized Sim.	Applies baseline rescaling to improve comparability across models.
CLIPScore [34]	Vision-Language	Measures similarity between image and caption embeddings in CLIP space.
SBERT Cosine [35]	Sentence-level	Computes cosine similarity between sentence embeddings.
Trainable Param.	Learnable Weights	Number of parameters updated during training.
Total Parameters	Model Size	Total parameters including frozen components.
Vision GFLOPs	Comp. Cost	Floating-point operations for the vision encoder.
Decoder GFLOPs	Comp. Cost	Floating-point operations for the Transformer decoder.
Total GFLOPs	Overall Complexity	Combined computational cost of vision and decoder modules.
GPU Latency	Performance	Average inference time (ms) per image measured on GPU.
CPU Latency	Performance	Average inference time (ms) per image measured on CPU.
Decode Speed	Gen. Speed	Number of tokens generated per second during decoding.
FPS	Throughput	Number of images processed per second.
Speed-up	Efficiency	Inference speed improvement relative to the BLIP baseline.

4.4 Baseline Methods

We compare Photon against representative lightweight and pretrained captioning approaches:

- **ClipCap** [1]: CLIP prefix conditioning with GPT-2 decoder.
- **CapDec** [5]: Decoder trained to reconstruct captions from CLIP embeddings.
- **SmallCap** [7]: Retrieval-augmented lightweight captioning.
- **BLIP** [8]: Large-scale vision-language pretraining model.
- **ViT-GPT2** [6]: Vision Transformer-based image encoder combined with a GPT-2 decoder for autoregressive image caption generation using projected visual features.

All reported baseline results are obtained from official implementations or publicly available pretrained models, evaluated on the MS-COCO Karpathy split using standard captioning metrics.

Fairness of Comparison. All models are evaluated on the same dataset split and using identical evaluation metrics to ensure consistent benchmarking. Photon is trained on the MS-COCO Karpathy training split, while baseline models are evaluated using their publicly available pretrained weights. We acknowledge that baseline methods differ in architecture, parameter scale, and training strategies, including the use of large pretrained decoders, retrieval augmentation, or cross-attention mechanisms. Therefore, the comparisons are intended to reflect practical performance and the trade-off between caption quality and computational efficiency, rather than strict equivalence in training conditions.

5 Results and Discussion

This section evaluates Photon across caption quality, efficiency, and generalization. We first present results on MS-COCO, followed by efficiency analysis and quality–efficiency trade-offs. We then assess cross-dataset performance, tokenizer robustness, and statistical significance, and conclude with deployment scalability through batch-level analysis.

5.1 Main Quantitative Results (MS-COCO)

This section presents quantitative comparisons with baseline methods and analyzes the performance of the proposed model across caption quality, semantic alignment, computational complexity, and deployment efficiency.

Table 2 reports standard captioning metrics on MS-COCO. **Photon (Ours)** attains strong overall caption quality, achieving the second-best performance in B@1, B@2, CIDEr, METEOR, and ROUGE-L. While ViT-GPT2 delivers the highest scores across all metrics, Photon demonstrates consistent competitive performance against established baselines such as BLIP and ClipCap.

To assess semantic alignment beyond n-gram overlap, Table 3 reports embedding-based metrics. Photon demonstrates competitive alignment with reference captions, indicating that the generated descriptions preserve scene-level semantics.

Table 2: Captioning performance comparison on the MS-COCO evaluation dataset. Best results are shown in **bold**, and the second-best results are indicated in *bold italics*.

Model	B@1 ↑	B@2 ↑	B@3 ↑	B@4 ↑	CIDEr ↑	SPICE ↑	METEOR ↑	ROUGE-L ↑
ClipCap	70.96	54.36	41.25	31.49	106.72	20.51	27.72	54.85
SmallCap	68.40	52.10	38.68	28.69	82.65	16.37	23.13	50.94
CapDec	68.31	50.64	36.86	26.78	92.37	18.42	25.20	51.29
BLIP	70.65	56.72	<i>44.45</i>	<i>34.36</i>	107.67	20.40	26.06	53.66
ViT-GPT2	77.73	61.80	47.06	35.39	119.06	21.15	27.94	57.11
Photon (Ours)	<i>74.65</i>	<i>58.03</i>	43.51	32.33	<i>108.59</i>	20.01	<i>26.87</i>	<i>55.06</i>

Table 3: Semantic alignment evaluation using BERTScore, CLIPScore, and SBERT similarity. Best results are shown in **bold**, and the second-best results are indicated in *bold italics*.

Model	BERTScore-F1 ↑	BERTScore-F1(Raw) ↑	CLIPScore ↑	SBERT Cosine ↑
ClipCap	65.27	<i>94.14</i>	75.39	73.88
SmallCap	58.29	92.96	68.56	64.93
CapDec	62.13	93.61	74.14	71.30
BLIP	59.83	93.22	72.51	72.99
ViT-GPT2	64.43	–	73.71	76.79
Photon (Ours)	65.63	94.20	<i>74.83</i>	<i>74.34</i>

5.2 Efficiency and Deployment Analysis

Model efficiency is summarized in Table 4. Photon achieves a favorable parameter and GFLOP profile relative to heavier baselines, supporting its design goal of deployment-oriented caption generation.

Table 4: Model complexity comparison in terms of parameters and computational cost (lower is better). For Photon, total GFLOPs (3.72) include the vision encoder, prefix projection module, and autoregressive decoding cost for generating a 15-token caption.

Model	Trainable Params ↓	Total Params ↓	Vision GFLOPs ↓	Decoder GFLOPs ↓	Total GFLOPs ↓
ClipCap	20M	230M	2.94	3.00	5.94
SmallCap	7M	257.61M	8.73	0.51	9.24
CapDec	2.9M	275M	19.66	5.02	24.68
BLIP	224M	224M	49.39	82.82	132.21
ViT-GPT2	239.20M	239.20M	–	–	17.33
Photon (Ours)	12.41M	33.91M	3.57	0.15	3.72

Runtime efficiency is evaluated on both GPU and CPU. Table 5 reports GPU-side inference characteristics, while Table 6 shows CPU latency comparisons, highlighting the practical advantages of the proposed lightweight decoding pipeline.

Table 5: GPU inference latency and throughput comparison. Speed-up is computed relative to BLIP.

Model	Vision (ms) ↓	Latency (ms/img) ↓	Decode (token/s) ↑	FPS ↑	Speed-Up ↑
ClipCap	9.75	156.77	80.38	6.38	1.69×
SmallCap	10.85	152.13	42.32	6.57	1.74×
CapDec	13.37	288.81	62.24	3.46	0.92×
BLIP	14.46	265.37	43.13	3.77	1.00×
ViT-GPT2	8.52	189.41	70.58	5.28	1.40×
Photon (Ours)	19.95	109.96	106.02	9.09	2.41×

Table 6: CPU inference latency and throughput comparison. Speed-up is computed relative to BLIP.

Model	Vision (ms) ↓	Latency (ms/img) ↓	Decode (token/s) ↑	FPS ↑	Speed-Up ↑
ClipCap	60.76	888.22	12.99	1.13	2.22×
SmallCap	53.42	384.84	17.29	2.60	5.11×
CapDec	172.53	2909.98	4.72	0.34	0.68×
BLIP	452.83	1967.99	6.12	0.51	1.00×
ViT-GPT2	114.61	853.54	19.06	1.17	2.31×
Photon (Ours)	113.08	237.67	84.63	4.21	8.28×

5.3 Quality–Efficiency Trade-off

To better understand the quality–efficiency characteristics of Photon, we analyze caption quality as a function of compute cost and runtime.

Figure 3 visualizes CIDEr versus decoder GFLOPs for Photon and baseline/variant models. Photon lies near the quality–compute frontier, achieving strong CIDEr with moderate computational cost. Notably, variants that increase capacity (e.g., larger prefix or deeper decoder) show diminishing CIDEr gains relative to increased compute, while smaller variants incur a visible quality drop.

Figure 4 plots CIDEr versus projection+decoder GFLOPs for the ablation configurations. The baseline configuration (L6/P8/D256) is highlighted with a star, and the dashed line denotes the Pareto frontier representing the best quality–compute trade-offs. Prefix variants are denoted as P1, P2, P4, and P16, corresponding to prefix token lengths of 1, 2, 4, and 16, respectively. Decoder depth variants are labeled L2, L4, and L8, indicating Transformer decoders with 2, 4, and 8 layers. Width variants are marked as D192 and D320, representing model hidden dimensions of 192 and 320. Conditioning strategies include Cross (cross-attention conditioning), Global (global feature conditioning), and Mean (mean-pooled feature conditioning). The FP32 marker indicates full 32-bit precision training, while other points use mixed-precision settings. Together, these configurations illustrate how architectural choices influence the trade-off between caption quality and computational cost.

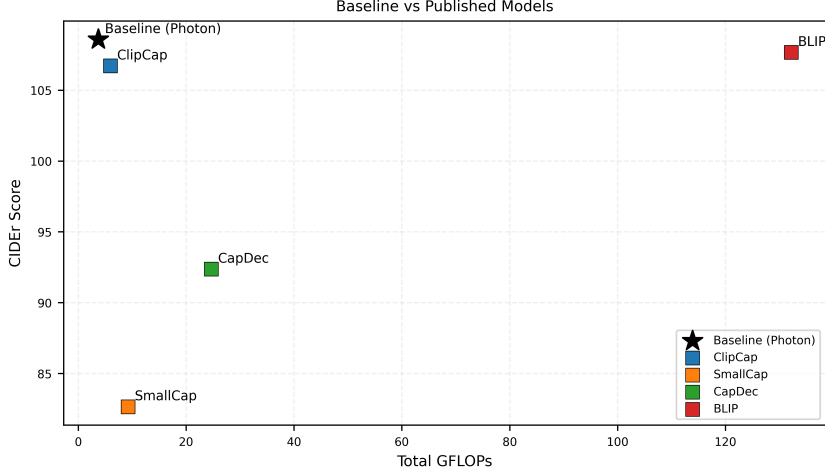


Fig. 3: CIDEr score versus total GFLOPs comparing Photon with established captioning models. The reported total GFLOPs account for the vision encoder forward pass, prefix projection, and full autoregressive decoding. Photon lies near the Pareto frontier, achieving strong caption quality at significantly reduced computational cost relative to heavier architectures, highlighting its efficiency-oriented design.

5.4 Cross-Dataset Generalization

To assess generalization beyond MS-COCO, we evaluate the trained model on multiple out-of-domain datasets, including Flickr30K [24], NoCaps [25], and TextCaps [26].

Tables 7 and 8 report captioning and semantic alignment metrics on Flickr30K test set. Similarly, Tables 9 and 10 present results on the NoCaps validation set, while Tables 11 and 12 summarize performance on the TextCaps validation set.

The results show that Photon consistently maintains competitive performance under varying caption distributions and domain shifts. In particular, Photon achieves strong performance in both n-gram based metrics (BLEU, CIDEr) and embedding-based metrics (BERTScore, CLIPScore, SBERT), demonstrating robust cross-dataset generalization.

Overall, these results indicate that prefix-conditioned visual grounding learned on MS-COCO transfers effectively to diverse datasets without requiring architectural modifications.

5.5 Tokenizer Generalization Analysis

To investigate whether training the tokenizer on MS-COCO captions limits cross-domain generalization, we conduct a comparative study using multiple tokenization strategies, including *CLIP tokenizer*, *COCO-trained BPE*, *CommonCrawl-BPE*, and *SentencePiece*. The evaluation is performed across both in-domain (MS-COCO Karpathy test split) and out-of-domain datasets, including Flickr30K, Flickr8k, NoCaps, and TextCaps.

Table 7: Flickr30k test set captioning performance. Best results are shown in **bold**, and the second-best results are indicated in ***bold italics***.

Method	B@1↑	B@2↑	B@3↑	B@4↑	CIDEr↑	SPICE↑	METEOR↑	ROUGE-L↑
BLIP	60.72	44.68	32.55	23.46	53.88	13.40	18.58	43.85
CapDec	56.62	36.72	23.27	14.72	31.53	11.03	17.31	40.18
ClipCap	59.29	39.18	25.45	16.44	41.52	12.01	18.65	41.89
SmallCap	59.99	41.46	27.76	18.50	37.07	10.80	16.90	42.46
ViT-GPT2	<i>62.89</i>	42.06	26.42	16.40	39.21	11.08	17.86	42.39
Photon (Ours)	64.72	44.81	<i>29.69</i>	<i>19.46</i>	<i>43.85</i>	<i>12.47</i>	19.20	44.70

Table 8: Flickr30k semantic evaluation. Best results are shown in **bold**, and the second-best results are indicated in ***bold italics***.

Method	BERTScore-F1↑	CLIPScore↑	SBERT↑
BLIP	52.10	68.88	<i>57.39</i>
CapDec	53.37	69.68	49.83
ClipCap	<i>56.10</i>	<i>72.71</i>	54.32
SmallCap	52.90	66.58	50.80
ViT-GPT2	52.94	68.79	50.73
Photon (Ours)	57.73	74.83	74.34

Table 9: NoCaps validation captioning performance. Best results are shown in **bold**, and the second-best results are indicated in ***bold italics***.

Method	B@1↑	B@2↑	B@3↑	B@4↑	CIDEr↑	SPICE↑	METEOR↑	ROUGE-L↑
BLIP	78.94	65.26	51.66	39.69	97.86	13.54	27.11	56.69
CapDec	65.39	46.13	31.30	21.06	44.22	8.16	19.99	45.57
ClipCap	71.59	53.55	38.42	27.12	<i>65.76</i>	<i>10.44</i>	<i>23.47</i>	50.30
SmallCap	65.63	50.11	36.55	25.82	45.82	7.81	19.62	48.00
ViT-GPT2	73.79	55.68	39.55	27.25	57.87	9.30	22.19	50.02
Photon (Ours)	<i>74.27</i>	<i>57.14</i>	<i>41.53</i>	<i>29.30</i>	63.44	10.07	23.15	<i>51.13</i>

Table 10: NoCaps semantic evaluation. Best results are shown in **bold**, and the second-best results are indicated in ***bold italics***.

Method	BERTScore-F1↑	CLIPScore↑	SBERT↑
BLIP	59.99	73.34	74.71
CapDec	53.15	65.02	57.71
ClipCap	58.55	<i>70.71</i>	<i>66.23</i>
SmallCap	52.49	63.64	60.49
ViT-GPT2	55.41	66.14	63.19
Photon (Ours)	<i>59.13</i>	68.49	65.80

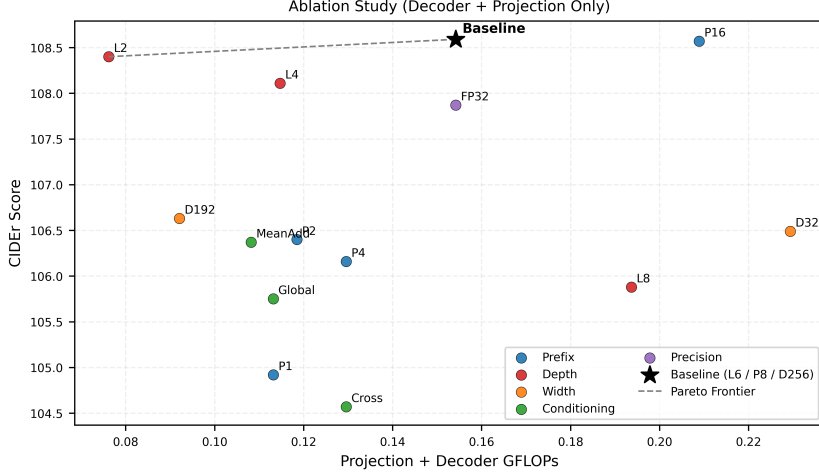


Fig. 4: CIDEr versus inference latency comparing Photon with baseline models. Photon achieves strong caption quality while maintaining competitive runtime performance.

Table 11: TextCaps validation captioning performance. Best results are shown in **bold**, and the second-best results are indicated in *bold italics*.

Method	B@1↑	B@2↑	B@3↑	B@4↑	CIDEr↑	SPICE↑	METEOR↑	ROUGE-L↑
BLIP	59.76	41.13	27.76	18.23	48.06	13.92	17.15	39.09
CapDec	48.07	27.20	15.23	8.59	19.97	7.77	12.58	31.27
ClipCap	53.56	33.32	20.31	12.35	<i>31.01</i>	<i>10.64</i>	<i>14.74</i>	34.44
SmallCap	46.29	29.33	18.38	11.09	21.41	8.28	12.21	32.71
ViT-GPT2	<i>55.26</i>	<i>35.46</i>	<i>21.44</i>	12.48	27.29	9.60	14.35	<i>35.25</i>
Photon (Ours)	55.29	35.00	21.26	<i>12.47</i>	28.84	9.11	14.43	34.71

As shown in Table 13, the **COCO-BPE tokenizer consistently achieves the best or highly competitive performance across all datasets and evaluation metrics**, including BLEU, CIDEr, and BERTScore-F1. In particular, on Flickr30K and NoCaps, COCO-BPE outperforms alternative tokenizers by a clear margin in CIDEr and BLEU scores, demonstrating strong cross-dataset generalization.

While SentencePiece achieves comparable performance in certain metrics such as METEOR, the overall results indicate that **COCO-BPE provides a well-balanced representation for captioning tasks**, effectively capturing the linguistic structure of image descriptions.

These results indicate that the COCO-trained BPE tokenizer generalizes effectively across multiple datasets, including out-of-domain benchmarks such as Flickr30k, NoCaps, and TextCaps. Despite being trained on MS-COCO captions, it consistently achieves competitive or superior performance compared to alternative tokenization

Table 12: TextCaps semantic evaluation. Best results are shown in **bold**, and the second-best results are indicated in *bold italics*.

Method	BERTScore-F1↑	CLIPScore↑	SBERT↑
BLIP	38.26	69.84	57.99
CapDec	34.49	60.61	43.83
ClipCap	38.35	66.81	51.10
SmallCap	33.82	60.79	46.81
ViT-GPT2	36.67	62.22	48.68
Photon (Ours)	38.68	64.20	50.69

strategies, suggesting that it captures broadly useful linguistic patterns rather than overfitting to a specific domain.

These findings suggest that **training the tokenizer on MS-COCO does not hinder generalization**, but instead aligns well with the captioning domain, enabling robust transfer to diverse datasets without requiring domain-specific tokenization. This directly addresses the concern regarding tokenizer-induced domain limitation.

Table 13: Tokenizer comparison across datasets. Best results are shown in **bold**. All metrics are higher-is-better (↑).

Dataset	Tokenizer	B@1↑	B@2↑	B@3↑	B@4↑	CIDEr↑	SPICE↑	METEOR↑	ROUGE-L↑	BERTScore-F1↑
COCO	CLIP	39.51	28.32	19.60	13.39	3.97	20.68	25.47	37.48	31.80
	CommonCrawl-BPE	56.62	39.25	25.78	16.47	53.49	15.33	20.89	39.70	30.85
	SentencePiece	73.52	56.93	42.50	31.39	106.55	20.20	27.02	54.89	65.19
	Photon (Ours)	74.43	57.72	43.17	31.96	108.07	20.18	26.95	54.94	65.44
Flickr30k	CLIP	36.51	23.06	14.09	8.57	3.95	13.51	19.17	32.68	27.31
	CommonCrawl-BPE	50.49	31.47	18.40	10.71	24.16	9.45	15.14	33.48	25.42
	SentencePiece	63.41	43.59	28.83	18.84	42.63	12.39	19.17	44.27	57.18
	COCO BPE	64.72	44.81	29.69	19.46	43.85	12.47	19.20	44.70	57.73
Flickr8k	CLIP	35.54	23.25	14.42	8.73	2.63	15.66	21.06	33.36	30.23
	CommonCrawl-BPE	49.83	31.82	19.11	11.19	29.47	11.47	17.19	35.88	26.12
	SentencePiece	63.90	45.30	30.20	19.72	54.09	15.08	21.85	47.82	55.70
	COCO BPE	64.80	46.11	30.84	20.21	55.37	15.15	21.92	48.21	56.09
NoCaps	CLIP	44.72	31.46	21.07	13.70	4.08	11.89	22.89	37.07	27.48
	CommonCrawl-BPE	60.31	42.29	27.31	17.04	33.96	7.94	18.76	39.03	27.44
	SentencePiece	73.80	56.57	41.00	28.86	62.52	10.10	23.25	51.37	58.89
	COCO BPE	74.27	57.14	41.53	29.30	63.44	10.07	23.15	51.13	59.13
TextCaps	CLIP	33.55	19.78	11.17	6.12	2.99	10.78	15.41	28.01	10.86
	CommonCrawl-BPE	46.64	27.51	15.10	8.04	17.37	8.04	12.60	28.90	13.44
	SentencePiece	54.94	34.75	21.05	12.32	28.26	9.84	14.46	34.90	37.94
	COCO BPE	55.29	35.00	21.26	12.47	28.84	9.91	14.43	34.71	38.68

5.6 Statistical Significance Analysis

To evaluate the stability and reliability of the proposed Photon model, we conduct each experiment over 30 independent runs and report the mean and standard deviation for all evaluation metrics. This enables us to assess the consistency of the model under different random initializations and training dynamics.

Since multiple-run statistics for baseline methods are not publicly available, we perform statistical testing using a one-sample t-test, where baseline results are treated

as fixed reference values. For each metric, the t-value is computed as:

$$t = \frac{\mu_{\text{Photon}} - x_{\text{baseline}}}{\sigma_{\text{Photon}}/\sqrt{n}}, \quad (13)$$

where μ_{Photon} and σ denote the mean and standard deviation of Photon over $n = 30$ runs, and x_{baseline} is the reported value of the baseline method.

The corresponding two-tailed p-value is computed as:

$$p = 2 \cdot (1 - F_t(|t|; n - 1)), \quad (14)$$

where $F_t(\cdot)$ denotes the cumulative distribution function of the Student’s t-distribution. The p-values are obtained using a two-tailed t-distribution with $n - 1$ degrees of freedom.

For clarity, t- and p-values in Tables 14, 15, and 16 are reported in the order of (B@4, CIDEr, METEOR).

Tables 14, 15, and 16 present the statistical comparison results on MS-COCO, Flickr30k, and TextCaps, respectively. Across all datasets, Photon exhibits consistently low standard deviation, indicating stable and reliable performance.

On MS-COCO (Table 14), Photon achieves competitive performance across all metrics. For example, in terms of CIDEr, Photon slightly outperforms BLIP, resulting in a t-value of 2.70 and a p-value of 0.011, indicating statistical significance at the 0.05 level. In contrast, for BLEU-4(B@4) and METEOR, the negative t-values against stronger baselines such as BLIP and ViT-GPT2 indicate that Photon underperforms on these metrics, while still maintaining very low variance. Comparisons with lighter baselines such as CapDec and SmallCap show large positive t-values, indicating that Photon significantly outperforms these methods.

On Flickr30k (Table 15), larger absolute t-values are observed across most comparisons. Photon achieves lower BLEU-4(B@4) and CIDEr scores compared to stronger baselines like BLIP, as reflected by large negative t-values. However, the consistently low standard deviation highlights that these results are stable across runs. At the same time, Photon outperforms weaker baselines such as CapDec and SmallCap, with large positive t-values and very small p-values.

On TextCaps (Table 16), a similar trend is observed. Photon underperforms compared to BLIP in BLEU-4(B@4) and CIDEr, as indicated by large negative t-values, but remains competitive against other lightweight baselines. For instance, the comparison with ClipCap shows a small t-value for BLEU-4(B@4) (2.17) with a p-value of 0.038, indicating a marginal but statistically significant improvement, while differences in CIDEr and METEOR remain significant.

Overall, the results highlight two key observations: (1) Photon demonstrates highly stable performance across multiple runs, as evidenced by consistently low standard deviation values, and (2) performance differences with baseline methods are statistically significant in most cases, as indicated by low p-values. However, these comparisons should be interpreted with caution, since baseline variance is not available. Thus, the statistical analysis primarily reflects the stability of Photon and its

relative differences with respect to fixed reference baselines, rather than a fully paired statistical comparison.

These findings confirm that the proposed prefix-conditioned architecture achieves a strong balance between efficiency, stability, and competitive performance across diverse datasets.

Table 14: Statistical comparison on MS-COCO. Photon (Ours) results: B@4 = 31.96, CIDEr = 108.07, METEOR = 26.95 (mean $\pm$ std over 30 runs). Baselines are treated as fixed reference values. t- and p-values are reported in the order of (B@4, CIDEr, METEOR).

Method	B@4	CIDEr	METEOR	t-value	p-value
BLIP	34.36	107.67	26.06	-37.9 / 2.70 / 10.3	< 0.001 / 0.011 / < 0.001
CapDec	26.78	92.37	25.20	89.2 / 106.7 / 50.4	< 0.001 / < 0.001 / < 0.001
ClipCap	31.49	106.72	27.72	7.54 / 11.3 / -29.5	< 0.001 / < 0.001 / < 0.001
SmallCap	28.69	82.65	23.13	56.0 / 164.0 / 109.3	< 0.001 / < 0.001 / < 0.001
ViT-GPT2	35.39	119.06	27.94	-55.2 / -74.2 / -38.0	< 0.001 / < 0.001 / < 0.001

Table 15: Statistical comparison on Flickr30k. Photon (Ours) results: B@4 = 19.46, CIDEr = 43.85, METEOR = 19.20 (mean $\pm$ std over 30 runs). t- and p-values are reported in the order of (B@4, CIDEr, METEOR).

Method	B@4	CIDEr	METEOR	t-value	p-value
BLIP	23.46	53.88	18.58	-54.2 / -62.0 / 24.1	< 0.001 / < 0.001 / < 0.001
CapDec	14.72	31.53	17.31	63.5 / 75.9 / 52.4	< 0.001 / < 0.001 / < 0.001
ClipCap	16.44	41.52	18.65	43.3 / 14.5 / 20.4	< 0.001 / < 0.001 / < 0.001
SmallCap	18.50	37.07	16.90	13.7 / 38.7 / 75.0	< 0.001 / < 0.001 / < 0.001

Table 16: Statistical comparison on TextCaps. Photon (Ours) results: B@4 = 12.47, CIDEr = 28.84, METEOR = 14.43 (mean $\pm$ std over 30 runs). t- and p-values are reported in the order of (B@4, CIDEr, METEOR).

Method	B@4	CIDEr	METEOR	t-value	p-value
BLIP	18.23	48.06	17.15	-101.5 / -249.0 / -72.0	< 0.001 / < 0.001 / < 0.001
CapDec	8.59	19.97	12.58	70.2 / 116.7 / 84.5	< 0.001 / < 0.001 / < 0.001
ClipCap	12.35	31.01	14.74	2.17 / -28.2 / -14.3	0.038 / < 0.001 / < 0.001
SmallCap	11.09	21.41	12.21	24.6 / 93.5 / 107.4	< 0.001 / < 0.001 / < 0.001

5.7 Batch-Scaling and Deployment Behavior

We further evaluate deployment behavior under batched inference to understand throughput scaling and memory usage.

Table 17 reports batch-wise latency and throughput measurements. Throughput increases substantially with batch size, while per-image latency decreases as preprocessing and kernel launch overheads are amortized.

Table 18 summarizes scaling behavior across batch sizes, including memory footprint.

Table 17: Inference latency breakdown across batch sizes (average caption length fixed to 15 tokens).

Batch	Preproc (ms) ↓	Vision (ms) ↓	Proj (ms) ↓	Dec (ms) ↓	Batch Total (ms) ↓	Per Img (ms) ↓
1	7.30	19.33	0.18	86.22	113.03	113.03
2	14.77	19.90	0.18	86.42	121.27	60.64
4	29.75	20.29	0.19	87.31	137.54	34.39
8	60.78	21.80	0.24	86.99	169.82	21.23
16	122.64	38.31	0.24	87.06	248.26	15.52
32	251.50	68.09	0.24	90.52	410.36	12.82
64	511.75	124.42	0.30	97.40	733.87	11.47
128	1059.23	247.97	0.47	109.44	1417.10	11.07
256	2047.41	511.29	0.54	104.20	2663.44	10.40
512	4140.42	994.32	0.60	92.81	5228.15	10.21

Table 18: Throughput and GPU memory scaling across batch sizes.

Batch	ms/token ↓	Tokens/s ↑	Peak GPU Mem (MB) ↓	Batches Measured
1	5.7478	173.99	394.20	1024
2	2.8806	347.15	404.35	512
4	1.4552	687.19	426.78	256
8	0.7249	1379.52	467.01	128
16	0.3627	2756.94	551.23	64
32	0.1886	5302.97	719.66	32
64	0.1015	9962.93	1056.49	16
128	0.0569	17890.07	1730.19	32
256	0.0271	37543.06	3077.59	16
512	0.0121	82769.87	5771.98	8

Figure 5(a) shows the linear-scale throughput behavior, while Figure 5(b) presents the same results using a logarithmic scale.

6 Ablation Study

To analyze the design choices of Photon, we conduct ablation studies on prefix length, decoder capacity, conditioning strategy, and precision settings. All experiments use the same training setup unless otherwise stated. We report caption quality metrics along with computational cost (GFLOPs) and decoding latency (TPOT). GFLOPs correspond to generating a 15-token caption, and TPOT denotes the mean per-token decoding time ($\mu \pm \sigma$) measured over 1000 runs.

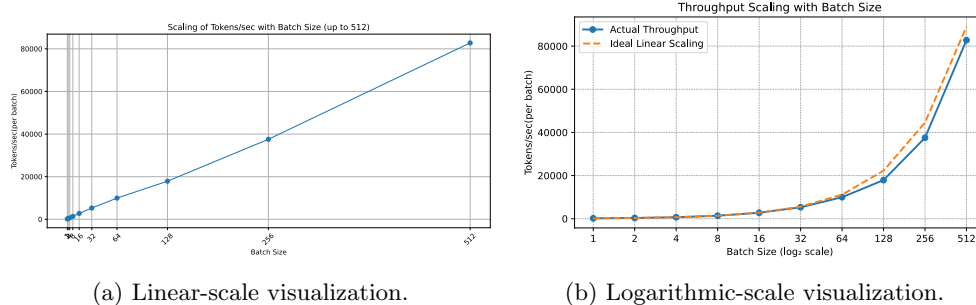


Fig. 5: Throughput scaling measured as batch-level decoding throughput (tokens/sec per batch) as a function of batch size. The reported values correspond to the total number of tokens generated per second across the entire batch. Throughput increases nearly linearly with batch size, reaching over 82,000 tokens/sec at batch size 512. Sub-figure (a) presents the results using a linear scale, where smaller batch sizes (1, 2, 4, and 8) appear closely clustered due to the large dynamic range of the axis. Sub-figure (b) shows the same results using a logarithmic scale, which improves the visibility of smaller batch sizes and highlights the consistent scaling behavior across the entire range. This behavior indicates efficient GPU utilization and strong parallel decoding capability under batched inference.

6.1 Effect of Prefix Length

We first examine the impact of varying the number of visual prefix tokens K . The prefix length is varied across $\{1, 2, 4, 8, 16\}$ while keeping the decoder depth and hidden dimension fixed.

Increasing the prefix length from 1 to 8 tokens consistently improves caption quality across CIDEr and BLEU-4, indicating that additional prefix tokens enhance the model’s ability to encode visual context. The best CIDEr performance is achieved at $K = 8$ as shown in Table 19.

Further increasing the prefix length to 16 yields negligible improvement in caption quality while significantly increasing parameter count and computational cost. This suggests diminishing returns beyond a moderate prefix capacity.

Importantly, inference latency remains largely stable across prefix configurations, indicating that prefix expansion primarily affects parameter size rather than decoding time.

These results demonstrate that $K = 8$ provides an effective balance between visual conditioning capacity and computational efficiency.

The choice of $K = 8$ prefix tokens is motivated by both empirical performance and representational efficiency. For smaller values ($K = 1-4$), the model lacks sufficient capacity to capture diverse scene semantics, resulting in reduced caption quality. Increasing K improves performance by providing richer visual conditioning; however, beyond $K = 8$, the gains saturate, as the global visual embedding already encodes compact semantic information. Moreover, larger K increases sequence length and the quadratic cost of self-attention, leading to higher computational overhead without

meaningful performance improvement. Therefore, $K = 8$ provides an effective balance between representation capacity and efficiency.

Table 19: Prefix length ablation study (baseline: 6 layers, $d = 256$, $K = 8$). This setup isolates the impact of prefix-token scaling on decoder-side quality–efficiency trade-offs.

Prefix	CIDEr $\uparrow$	B@4 $\uparrow$	METEOR $\uparrow$	SPICE $\uparrow$	Params (M) $\downarrow$	GFLOPs $\downarrow$	Latency ($\mu \pm \sigma$) (ms) $\downarrow$
1	104.92	30.78	26.52	19.56	7.36	0.1132	6.47 ± 0.055
2	106.40	31.14	26.77	19.91	7.69	0.1185	6.50 ± 0.070
4	106.16	31.13	26.87	20.03	8.73	0.1296	6.45 ± 0.052
8 (Baseline)	108.59	32.33	26.87	20.01	12.41	0.1542	6.45 ± 0.067
16	108.57	32.04	26.89	20.36	26.04	0.2089	6.42 ± 0.042

6.2 Decoder Depth Analysis

Next, we evaluate the effect of decoder depth while fixing the hidden dimension and prefix length. The number of Transformer layers is varied across $\{2, 4, 6, 8\}$.

Increasing depth from 2 to 6 layers yields consistent improvements in CIDEr and BLEU-4, demonstrating that deeper autoregressive modeling enhances language generation quality. However, moving from 6 to 8 layers provides only marginal gains while increasing computational cost and latency as shown in Table 20.

This behavior suggests that a 6-layer configuration is sufficient to capture the required multimodal dependencies for caption generation under the prefix-conditioned setup.

Table 20: Decoder depth ablation with $d = 256$ and $K = 8$.

Layers	CIDEr $\uparrow$	B@4 $\uparrow$	Params (M) $\downarrow$	GFLOPs $\downarrow$	Latency ($\mu \pm \sigma$) (ms) $\downarrow$
2	108.40	31.88	9.0	0.0762	2.30 ± 0.023
4	108.11	31.75	10.7	0.1147	4.38 ± 0.058
6	108.59	32.33	12.41	0.1542	6.45 ± 0.067
8	105.88	31.45	14.11	0.1937	8.58 ± 0.060

6.3 Decoder Width Analysis

We further investigate the impact of hidden dimension d while keeping depth and prefix length fixed. Increasing d improves representational capacity and slightly enhances caption quality as shown in Table 21. However, gains beyond the baseline configuration are modest relative to the increase in parameter count and GFLOPs.

These findings indicate that moderate-width decoders are adequate when visual information is efficiently injected through prefix tokens.

Table 21: Decoder width ablation with 6 layers and $K = 8$.

Hidden Dim	CIDEr $\uparrow$	B@4 $\uparrow$	Params (M) $\downarrow$	GFLOPs $\downarrow$
192	106.63	31.75	7.56	0.0921
256	108.59	32.33	12.41	0.1542
320	106.49	31.68	18.42	0.2294

6.4 Conditioning Strategy Comparison

To assess the effectiveness of prefix-only conditioning, we compare it against alternative strategies, including:

- Single-token prefix conditioning,
- Mean-pooled conditioning (adding the visual embedding to each token),
- Lightweight cross-attention.

Prefix-only conditioning consistently outperforms single-token and mean-pooled approaches, confirming that distributing visual information across multiple prefix tokens improves representational flexibility as shown in Table 22. While cross-attention variants may yield marginal improvements, they incur additional computational overhead. The prefix-only approach achieves a favorable balance between performance and efficiency without requiring architectural modifications to the decoder.

Table 22: Comparison of visual conditioning strategies.

Method	CIDEr $\uparrow$	B@4 $\uparrow$	Params (M) $\downarrow$	GFLOPs $\downarrow$	Latency ($\mu \pm \sigma$) (ms) $\downarrow$
Prefix (Baseline)	108.59	32.33	12.41	0.1542	6.45 $\pm$ 0.067
Global Token	105.75	30.96	7.36	0.1132	6.47 $\pm$ 0.031
Mean Add	106.37	31.59	7.36	0.1082	6.52 $\pm$ 0.093
Cross-Attn	104.57	30.74	10.31	0.1296	6.46 $\pm$ 0.042

6.5 Precision and Deployment Settings

Finally, we analyze the impact of mixed-precision inference. Using FP16 decoding with FP32 projection results in negligible changes in caption quality compared to full FP32 inference, as shown in Table 23. The measured decoding latency remains nearly identical between the two settings, differing by only 0.02ms on average across 1000 runs.

These results indicate that Photon maintains stable captioning performance under mixed-precision deployment settings while preserving similar runtime characteristics, making it suitable for hardware-efficient inference without sacrificing caption quality.

Table 23: Impact of mixed precision on caption quality and latency.

Precision	CIDEr $\uparrow$	B@4 $\uparrow$	METEOR $\uparrow$	SPICE $\uparrow$	Latency ($\mu \pm \sigma$) (ms) $\downarrow$
Mixed (Default)	108.59	32.33	26.87	20.01	6.45 $\pm$ 0.067
Full FP32	107.87	31.85	26.99	20.27	6.47 $\pm$ 0.036

6.6 Discussion

Summary of Ablation Findings. Across all experiments: (i) prefix length improves performance up to $K = 8$ with diminishing returns beyond, (ii) decoder depth is effective up to 6 layers, (iii) increasing width provides limited benefit relative to cost, and (iv) prefix-based conditioning achieves a favorable balance compared to alternative strategies.

These findings validate the design choice of a compact architecture ($K = 8$, 6 layers, moderate width) that balances efficiency and performance.

7 Qualitative Analysis

We conduct qualitative comparisons to analyze semantic correctness, object grounding, and descriptive specificity across models.

Table 24A and Table 24B present representative examples comparing Ground Truth, ClipCap, BLIP, and the proposed Photon model.

Across diverse scene categories, Photon generates captions that remain semantically aligned with reference descriptions while preserving conciseness. In multiple cases, Photon correctly captures object categories, counts, and contextual relationships that are either omitted or less specific in baseline outputs. Compared to ClipCap and BLIP, Photon demonstrates improved consistency in scene-level interpretation and reduced semantic drift.

Despite strong overall behavior, limitations persist in visually complex or ambiguous scenarios. Table 25 presents representative failure cases.

Observed errors include misinterpretation of fine-grained activities, omission of subtle attributes, and occasional inaccuracies under cluttered or low-contrast conditions. These cases suggest that reliance on global visual embeddings may limit fine-grained spatial reasoning.

7.1 Error Analysis Discussion

To better understand the limitations of the proposed Photon model, we perform a qualitative error analysis using selected examples from the MS-COCO dataset. The analysis is based on manually inspected cases categorized into object-level errors, relation-level errors, and semantic inconsistencies.

Object-level errors: In some cases, Photon fails to correctly identify key objects. For example, in one instance where the ground truth describes “a cat laying on a pillow,” the model incorrectly generates “a cat sitting on top of a laptop.” This indicates confusion in recognizing fine-grained object details or contextual cues.

Relation-level errors: Photon occasionally captures the correct objects but fails to model relationships accurately. For instance, in images involving human-object interactions, the model may generate descriptions such as “a woman holding scissors” without capturing the intended relation “cutting her hair.” This suggests limitations in modeling spatial and action relationships.

Semantic drift and hallucination: More challenging cases reveal semantic inconsistencies, where the generated caption deviates significantly from the scene. For example, an image of a parked truck is incorrectly described as “a kite flying over a car on a beach.” Such errors indicate hallucination effects due to over-reliance on global representations.

Strengths: Despite these limitations, Photon demonstrates strong performance in capturing global scene semantics. In many cases, it produces more complete and contextually accurate descriptions than baseline models. For example, in images involving activities such as “a woman blow drying her hair,” Photon generates precise and complete captions, while baseline methods often produce incomplete or generic descriptions.

Overall, the error patterns suggest that while prefix-based conditioning enables efficient global understanding, it may limit fine-grained object grounding and relational reasoning. Future work can address these limitations by incorporating mechanisms for enhanced object-level alignment and relational modeling.

8 Limitations

Despite its strong quality–efficiency trade-off, Photon has some limitations. First, the model relies on a single global visual embedding expanded into prefix tokens. While efficient, this design may limit fine-grained spatial reasoning compared to region-based or cross-attention architectures that explicitly model object-level interactions.

Second, training and evaluation are primarily conducted on MS-COCO, with additional validation on Flickr30K. Although cross-dataset results are encouraging, broader generalization to domain-specific, low-resource, or out-of-distribution settings remains to be systematically validated.

Third, while Photon is competitive among lightweight models, it does not aim to match the reasoning capacity of very large multimodal foundation models. In scenarios requiring complex compositional reasoning or highly detailed scene descriptions, larger-scale architectures may still achieve superior performance.

9 Conclusion

We introduced Photon, a lightweight prefix-conditioned multimodal captioning framework that combines a frozen MobileCLIP encoder with a compact decoder-only Transformer incorporating modern architectural components.

Extensive evaluation on MS-COCO demonstrates that Photon achieves competitive caption quality, attaining a CIDEr score of 108.59 while requiring only 12.41M trainable parameters and 3.72 GFLOPs. The model matches or surpasses comparable lightweight baselines and approaches the performance of larger pretrained vision–language models, while offering substantial improvements in computational

efficiency. Inference experiments further show significant GPU and CPU speed-ups, confirming its suitability for deployment-oriented scenarios.

Qualitative analysis and ablation studies validate the effectiveness of prefix-based visual conditioning combined with modern decoder design choices. The results indicate that carefully calibrated lightweight architectures can deliver strong semantic fidelity without relying on heavy cross-attention mechanisms or billion-scale multimodal pretraining.

Overall, Photon establishes a favorable quality–efficiency trade-off, highlighting the potential of compact prefix-conditioned frameworks for practical image captioning applications.

10 Future Work

Several directions can extend the proposed framework. First, the prefix-based conditioning paradigm can be expanded to video captioning by integrating lightweight spatio-temporal encoders, enabling unified multimodal generation across images and short video clips.

Second, a more systematic theoretical and empirical investigation of prefix token capacity could provide deeper insight into the trade-off between visual information injection and language modeling complexity, potentially guiding principled architectural scaling.

Third, hardware-aware optimizations such as structured pruning, quantization, and compiler-level acceleration can be explored to further enhance real-time and edge deployment performance while maintaining caption quality.

Fourth, while this work adopts greedy decoding to ensure consistent and reproducible evaluation of model efficiency, advanced decoding strategies such as speculative decoding or assisted generation can be explored to further accelerate inference. Given the lightweight autoregressive design of Photon, integrating such techniques is expected to yield additional performance gains and improved latency.

References

- [1] Mokady, R., Hertz, A., Bermano, A.H.: ClipCap: CLIP Prefix for Image Captioning. Preprint at <https://arxiv.org/abs/2111.09734> (2021)
- [2] Su, J., Lu, Y., Pan, S., Wen, B., Liu, Y.: RoFormer: Enhanced transformer with rotary position embedding. In: Proceedings of the AAAI Conference on Artificial Intelligence (2021)
- [3] Chowdhery, A., Narang, S., Devlin, J., *et al.*: PaLM: Scaling language modeling with pathways. *Journal of Machine Learning Research* **24**(240), 1–113 (2023)
- [4] Anderson, P., He, X., Buehler, C., Teney, D., Johnson, M., Gould, S., Zhang, L.: Bottom-up and top-down attention for image captioning and visual question answering. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 6077–6086 (2018)

- [5] Nukrai, D., Mokady, R., Globerson, A.: Text-only training for image captioning using noise-injected CLIP. In: Findings of the Association for Computational Linguistics: EMNLP 2022, pp. 4055–4063 (2022)
- [6] Hao, J., Liu, H., Zhang, C.: ViT-GPT2 Image Captioning Model. Technical report at <https://www.haojiadong.com/report/AI/git-vit-imageCaptioning.pdf> (2023)
- [7] Ramos, R., Martins, B., Elliott, D., Kementchedjhieva, Y.: Smallcap: Lightweight image captioning prompted with retrieval augmentation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 2840–2849 (2023)
- [8] Li, J., Li, D., Xiong, C., Hoi, S.: Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In: Proceedings of the 39th International Conference on Machine Learning (ICML), pp. 12888–12900 (2022)
- [9] Zhang, P., Li, X., Hu, X., Yang, J., Zhang, L., Wang, L., Choi, Y., Gao, J.: Vinvl: Revisiting visual representations in vision-language models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 5579–5588 (2021)
- [10] Li, X., Yin, X., Li, C., Zhang, P., Hu, X., Zhang, L., Wang, L., Hu, H., Dong, L., Wei, F., *et al.*: Oscar: Object-semantics aligned pre-training for vision-language tasks. In: Proceedings of the European Conference on Computer Vision (ECCV), pp. 121–137 (2020)
- [11] Cao, L., Chen, H., Fan, X., Gama, J., Ong, Y.-S., Kumar, V.: Bayesian Federated Learning: A Survey. Preprint at <https://arxiv.org/abs/2304.13267> (2023)
- [12] Chen, H., Liu, H., Wu, Z., Fan, X., Cao, L.: FedSI: Federated subnetwork inference for efficient uncertainty quantification. *IEEE Transactions on Neural Networks and Learning Systems* **36**(9), 16975–16989 (2025) <https://doi.org/10.1109/TNNLS.2025.3562164>
- [13] Tian, Y., Chen, H., Zhu, X., Yu, X.: FedKT: A nature-inspired bayesian federated learning framework for privacy-preserving and lightweight consumer devices. *IEEE Transactions on Consumer Electronics*, 1–1 (2026) <https://doi.org/10.1109/TCE.2026.3667765>
- [14] Li, J., Shen, Y., Zhou, W., Chen, H.: Searchexpert: A genai-driven framework for reasoning-intensive multimedia information fusion through fine-tuning and reinforcement learning. *Information Fusion* **126**, 103665 (2026) <https://doi.org/10.1016/j.inffus.2025.103665>
- [15] Vasu, P.K.A., Pouransari, H., Faghri, F., Vemulapalli, R., Tuzel, O.: Mobileclip:

- Fast image-text models through multi-modal reinforced training. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 15963–15974 (2024)
- [16] Vasu, P.K.A., Gabriel, J., Zhu, J., Tuzel, O., Ranjan, A.: Fastvit: A fast hybrid vision transformer using structural reparameterization. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 5785–5795 (2023)
 - [17] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I.: Attention is all you need. In: Advances in Neural Information Processing Systems (NeurIPS), vol. 30 (2017)
 - [18] Zhang, B., Sennrich, R.: Root mean square layer normalization. In: Advances in Neural Information Processing Systems (NeurIPS), vol. 32 (2019)
 - [19] Dao, T., Fu, D., Ermon, S., Rudra, A., Ré, C.: FlashAttention: Fast and memory-efficient exact attention with io-awareness. In: Advances in Neural Information Processing Systems (NeurIPS), vol. 35, pp. 16344–16359 (2022)
 - [20] Williams, R.J., Zipser, D.: A learning algorithm for continually running fully recurrent neural networks. *Neural Computation* **1**(2), 270–280 (1989)
 - [21] Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L.: Microsoft COCO: Common objects in context. In: Proceedings of the European Conference on Computer Vision (ECCV), pp. 740–755 (2014)
 - [22] Karpathy, A., Fei-Fei, L.: Deep visual-semantic alignments for generating image descriptions. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 3128–3137 (2015)
 - [23] Sennrich, R., Haddow, B., Birch, A.: Neural machine translation of rare words with subword units. In: Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 1715–1725 (2016)
 - [24] Plummer, B.A., Wang, L., Cervantes, C.M., Caicedo, J.C., Hockenmaier, J., Lazebnik, S.: Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to-sentence models. In: Proceedings of the IEEE International Conference on Computer Vision (ICCV), pp. 2641–2649 (2015)
 - [25] Agrawal, H., Desai, K., Wang, Y., Chen, X., Jain, R., Johnson, M., Batra, D., Parikh, D., Lee, S., Anderson, P.: Nocaps: Novel object captioning at scale. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), pp. 8948–8957 (2019)
 - [26] Sidorov, O., Hu, R., Rohrbach, M., Singh, A.: Textcaps: A dataset for image captioning with reading comprehension. In: Proceedings of the European Conference

- on Computer Vision (ECCV), pp. 742–758 (2020)
- [27] Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. In: International Conference on Learning Representations (ICLR) (2019)
 - [28] Papineni, K., Roukos, S., Ward, T., Zhu, W.-J.: BLEU: A method for automatic evaluation of machine translation. In: Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 311–318 (2002)
 - [29] Vedantam, R., Zitnick, C.L., Parikh, D.: CIDEr: Consensus-based image description evaluation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4566–4575 (2015)
 - [30] Banerjee, S., Lavie, A.: METEOR: An automatic metric for mt evaluation with improved correlation with human judgments. In: Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures, pp. 65–72 (2005)
 - [31] Lin, C.-Y.: ROUGE: A package for automatic evaluation of summaries. In: Proceedings of the ACL Workshop on Text Summarization Branches Out, pp. 74–81 (2004)
 - [32] Anderson, P., Fernando, B., Johnson, M., Gould, S.: SPICE: Semantic propositional image caption evaluation. In: Proceedings of the European Conference on Computer Vision (ECCV), pp. 382–398 (2016)
 - [33] Zhang, T., Kishore, V., Wu, F., Weinberger, K.Q., Artzi, Y.: BERTScore: Evaluating text generation with bert. In: Proceedings of the International Conference on Learning Representations (ICLR) (2020)
 - [34] Hessel, J., Holtzman, A., Forbes, M., Le Bras, R., Choi, Y.: CLIPScore: A reference-free evaluation metric for image captioning. In: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 7514–7528 (2021)
 - [35] Reimers, N., Gurevych, I.: Sentence-bert: Sentence embeddings using siamese bert-networks. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP), pp. 3982–3992 (2019)

Table 24A: Qualitative comparison on representative natural and outdoor scenes, including animals and open environments. Ground-truth captions (GT1–GT5) are compared with captions generated by ClipCap, BLIP, and Photon (ours).

Image	Caption Comparison
	GT1: A flock of small birds flying in the sky over the water. GT2: Several birds that are flying together over a body of water. GT3: A flock of birds flying over a field. GT4: A black and white image showing birds flying over a body of water. GT5: A group of birds flying over the water ClipCap: A flock of birds flying over a body of water. BLIP: two birds flying in the sky over a body of water Photon (Ours): A flock of birds flying over a body of water.
	GT1: A herd of zebras standing together in a field. GT2: Herd of zebras standing on a rocky field. GT3: A herd of zebras standing in a dirt field. GT4: A group of zebras stand together on the rocky ground. GT5: a close up of a number of zebras on a dirt ground ClipCap: A herd of zebra standing next to each other on a dirt field. BLIP: the zebras are black and white Photon (Ours): A herd of zebras standing in a dirt field.
	GT1: A man holding a kite at the park GT2: A man that is standing in the grass with a kite. GT3: a man that is holding a large red and black kite GT4: A man holding a kite in a park GT5: a man holding onto a red and black kite in the park ClipCap: A man standing in a field holding a kite. BLIP: a man standing in a field holding a red and black kite Photon (Ours): A man holding a kite in a park.
	GT1: A lone man sitting on a bench on hill next to a large tree GT2: A man sitting on a bench by a tree. GT3: A person is sitting on a bench under a tree. GT4: A MAN IS SITTING ALONE ON A PARK BENCH GT5: Man contemplating his life while sitting on a bench under a tree. ClipCap: A man sitting on top of a wooden bench under a tree. BLIP: a person sitting on a bench under a tree Photon (Ours): A man sitting on a bench in a park.

Table 24B: Qualitative comparison on representative urban and daily-life scenes, including human activities and street environments. Ground-truth captions (GT1–GT5) are compared with captions generated by ClipCap, BLIP, and Photon (ours).

Image	Caption Comparison
	GT1: A person makes a sandwich on a paper plate. GT2: A plate that has food on it on a table. GT3: A close up picture of someone preparing a sandwich. GT4: A woman is preparing a sandwich on a paper plate. GT5: A person prepares a sandwich on a plate. ClipCap: A little girl eating a hot dog with ketchup and mustard. BLIP: a woman is putting a piece of food on a plate Photon (Ours): A woman is preparing a sandwich on a plate.
	GT1: The glaring lights of cars and street lights. GT2: a car is barely seen in the picture because it went by fast GT3: The car looks like a blur as it speeds past. GT4: A town street at night is lit up with neon lights. GT5: An image that is capturing headlights in motion. ClipCap: A red fire hydrant sitting on the side of a road. BLIP: a fire hydrant on the side of the road Photon (Ours): A car driving down a street at night.
	GT1: A red double decker bus driving down a street. GT2: a double stacked city bus coming down the street GT3: A double-decker bus on a busy street on an overcast day. GT4: A double decker bus driving down a street. GT5: Red, modern double decker bus driving down the street ClipCap: A red double decker bus driving down a street. BLIP: a red double decker bus driving down a city street Photon (Ours): A red double decker bus driving down a street.
	GT1: A group of people standing outside of a food truck GT2: some people are standing in front of a red food truck GT3: There are some people standing around a food truck GT4: A food truck parked outside at lunch time. GT5: A red food truck has a crowd of people by it. ClipCap: A group of people standing outside of a food truck. BLIP: a group of people standing in front of a food truck Photon (Ours): A group of people standing around a food truck.

Table 25: Failure cases highlighting model limitations.

Image	Caption Comparison
	GT1: Two well dressed standing together while on writes a note. GT2: Two men standing wearing tuxedos and one man writing something. GT3: a couple of men wearing suits and bow ties GT4: two males and they are wearing suits and bow ties GT5: A man in a tuxedo signs an autograph for another man ClipCap: Two men in suits looking at a piece of paper. BLIP: the man is wearing a suit Photon (Ours): A man in a suit and tie looking at his cell phone.
	GT1: A street post with a one way sign and street name sign along with holiday decorations. GT2: A 6th Street light pole is decorated for Christmas. GT3: Street sign decorated with bell shaped garland for Christmas. GT4: A light post with a Christmas decoration hanging from it. GT5: A Christmas decoration is on a light post. ClipCap: A green street sign hanging from the side of a traffic light. BLIP: a street sign with a wreath on top of it Photon (Ours): A street sign with a heart on it.
	GT1: A black cat laying next to a box of donuts. GT2: The cat is inside the house, poised next to the box of chocolate doughnuts. GT3: A black cat guards what's left of a box of donuts. GT4: a black cat laying next to a box of donuts GT5: A black cat sitting next to a box of doughnuts. ClipCap: A black cat sitting on top of a wooden floor next to a pair of shoes. BLIP: the cat is black Photon (Ours): A black cat sitting on a wooden table next to a box of donuts.